

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, C++, SQL, Javascript

Machine Learning: Audio/Music ML, Speech Applications, Deep Learning, Data Augmentation, Model Optimization, Digital Signal Processing (DSP), Real-time Systems

Frameworks & Tools: PyTorch, TensorFlow, Hugging Face, TensorRT, Docker, Kubernetes, FastAPI, AWS SageMaker

Data Engineering & Cloud: ETL Pipelines, PySpark, Airflow, AWS, GCP

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** for specialized applications, significantly improving **model performance** and training efficiency.
- Explored novel **ML architectures** and developed custom loss functions to enhance **model accuracy** and stability.
- Collaborated with **machine learning researchers** on advanced model development, contributing to academic publications.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Engineered and deployed **production software applications** leveraging **AI algorithms** to improve vitiligo segmentation accuracy by 94% and speed by 5x; integrated with **Docker** and **FastAPI** for client-facing solutions.
- Developed a **real-time system** for object detection, optimizing **ML model performance** for on-device deployment by reducing compute cost and time using **TensorRT**.
- Managed **MLOps pipelines** on **AWS SageMaker**, addressing **YOLO model** data drift and maintaining **Kubernetes infrastructure** for ML services, contributing to full feature development lifecycle.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed full-stack **production software applications** providing **AI-based services**, integrating machine learning functionality into web platforms.
- Built an **AI chatbot application** using the OpenAI API and relevant LLM frameworks, focusing on delivering **intelligent assistance**.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented **data augmentation pipelines** and ETL workflows for diverse structured and unstructured data, ensuring high-quality inputs for **model training pipelines**.

Projects

CAFA 6: Protein Function Prediction 2022

Deep Learning Competition

- Architected an ensemble ML model, leveraging **AlphaFold** for feature extraction and advanced techniques to improve protein function prediction accuracy.
- Enhanced **ML model performance** by 4% through custom feature engineering, demonstrating strong analytical and optimization skills.

Razanlytics: Real-time Bitcoin Price Forecasting 2022

Thesis Project

- Developed a **real-time forecasting system** for Bitcoin prices, implementing **model training pipelines** and evaluation tools for time-series ML models.
- Applied advanced time-series analysis and deep learning techniques to achieve robust predictive **model performance**.

Saaz: AI Self-Playing Chitralli Sitar 2023

Signal Processing, Music Generation

- Collaborated on the world's first **AI-driven** self-playing Sitar, integrating **real-time signal processing software** and **music generation LLMs** for creative output.
- Project featured on Coke Studio, showcasing successful application of **AI-driven music creation** technologies in a **consumer-facing application** context.

Hugging Face Open-Source

Contributor

- Co-authored technical documentation and implementation for **Zero-shot Image Classification models** on the Hugging Face Hub, demonstrating expertise in ML concepts.

Education

Friedrich Alexander Universität Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python